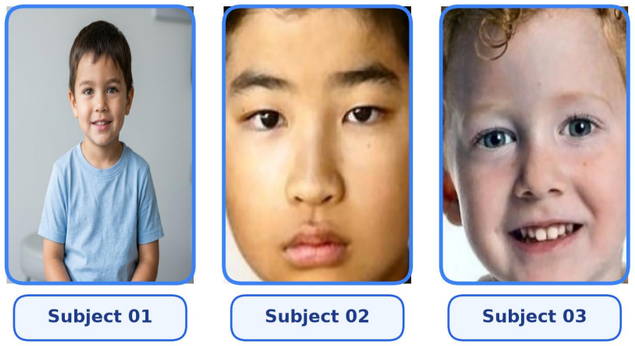
CNN-Based Autism Spectrum Disorder (ASD) Detection Pipeline

Reverse-C Sequential Flow: Data Ingestion → Facial Preprocessing → Dataset Splitting → CNN Architecture Comparison → Clinical Evaluation

01 Data Collection

Diverse Multi-Subject Pediatric Cohort

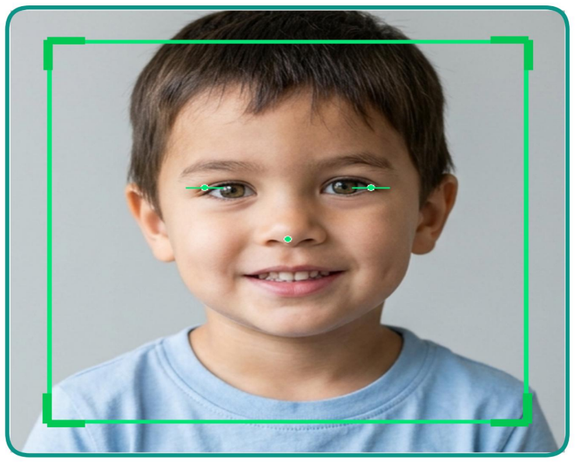


Subject 01 | Subject 02 | Subject 03

Pediatric Facial Dataset: [ ] Images

Raw Photos →

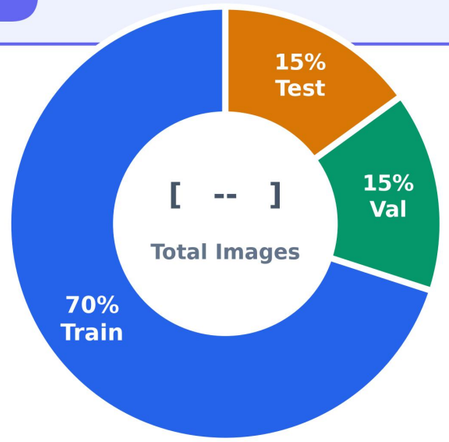
02 Face Preprocessing



MTCNN Detect & Align (224×224)

Aligned 224×224 →

03 Dataset Splitting



Train (70%)	Val (15%)	Test (15%)
[--] imgs	[--] imgs	[--] imgs

Stratified 70/15/15 Splits ↓

04 Model Training & Comparison

6 CNN CANDIDATE ARCHITECTURES (CROSS-ENTROPY LOSS, ADAM)

ResNet-50	Pending	[-- %]	Deep Residual (50-Layer)
ResNet-34	Pending	[-- %]	Residual Baseline (34-Layer)
VGG-19	Pending	[-- %]	Deep Feedforward (19-Layer)
VGG-16	Pending	[-- %]	Standard Baseline (16-Layer)
MobileNet-V2	Pending	[-- %]	Lightweight Inverted Residuals
AlexNet	Pending	[-- %]	Classic Convolutional Network

Optimal Backbone: [To Be Selected Based on Validation Results]

← Selected Backbone

05 Evaluation & Diagnosis

CLINICAL SCREENING OUTCOME

Binary Classification: ASD vs. Neurotypical Control

Predicted Label: [ ] | Model Confidence: [%]

Overall Test Accuracy	To Be Measured	[-- %]
AUROC Discriminability	To Be Measured	[-- %]
Diagnostic F1-Score	To Be Measured	[-- %]
Clinical Sensitivity / Specificity	To Be Measured	[-- / --]

Clinical Decision Support Ready • Target Inference Latency: [ ] ms